



JKBOSE 2026

CLASS 11TH

CONIC SECTIONS

MATHS

LECTURE-3

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TOPICS TO BE COVERED

1

Parabola

2

Related Problems

Parabola

For a Conic section, $\frac{PF}{PM} = e$ (constant)

For a Parabola, $e = 1$.

$$\Rightarrow \frac{PF}{PM} = 1 \Rightarrow PF = PM$$

* The locus of a point $P(x, y)$ in a plane such that its distance from a fixed point is equal to its distance from a fixed straight line, is known as a Parabola.

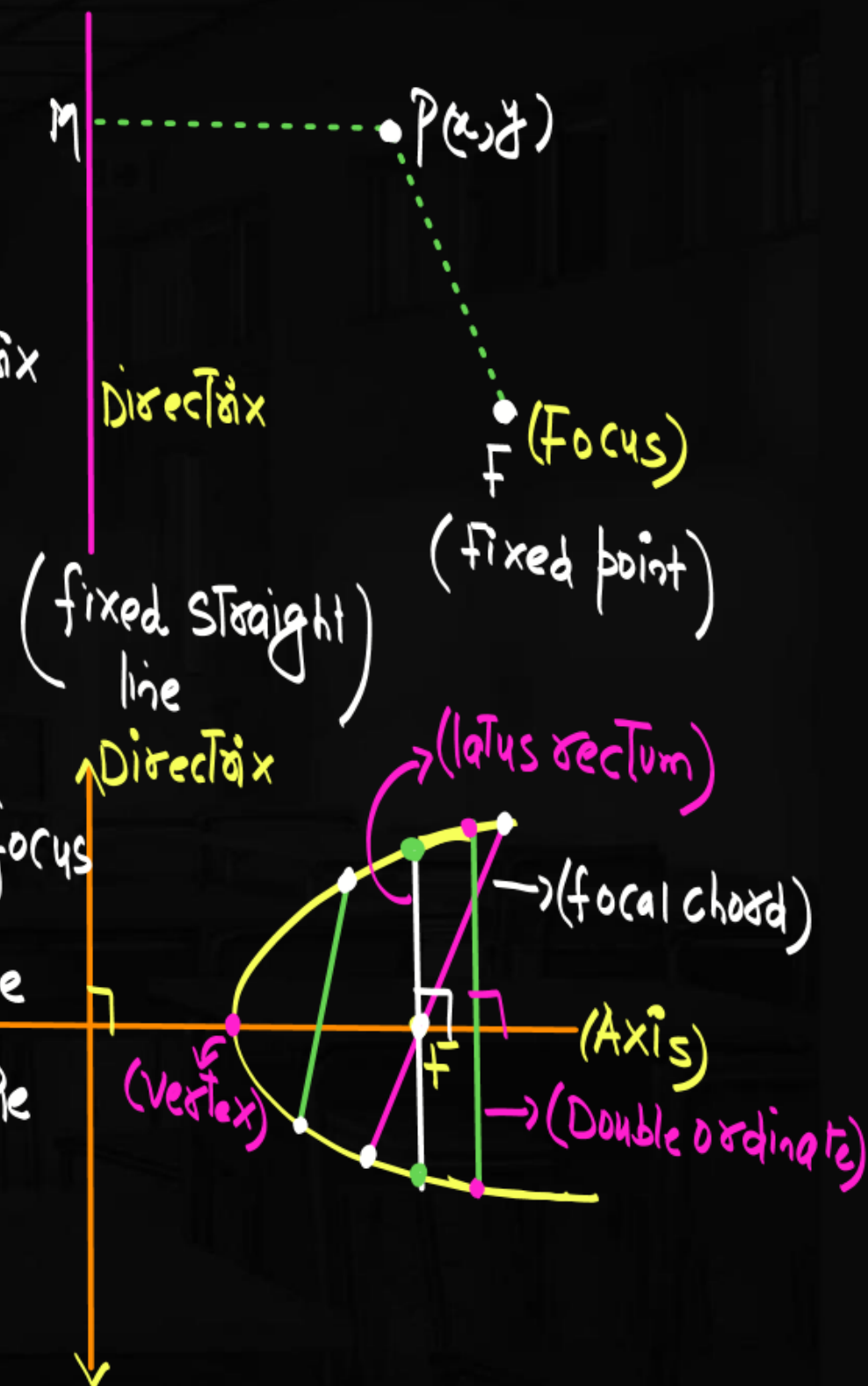
* The fixed point is known as the **focus** of the **parabola**.

* The fixed straight line is known as the **Directrix** of the parabola.

Terms Related to Parabola:

Axis of the Parabola:

Line passing through the focus & is perpendicular to the directrix is known as the axis of the parabola.



* **Vertex of Parabola**: The point of intersection of the parabola and the axis of the parabola is known as the vertex of the parabola.

* **Chord of the Parabola**: Line segment joining any two points on the parabola is known as the chord of the parabola.

* **Focal chord**: Chord passing through the focus of the parabola is known as the focal chord of the parabola.

* **Double ordinate**: Chord perpendicular to the axis of the parabola is known as Double ordinate.

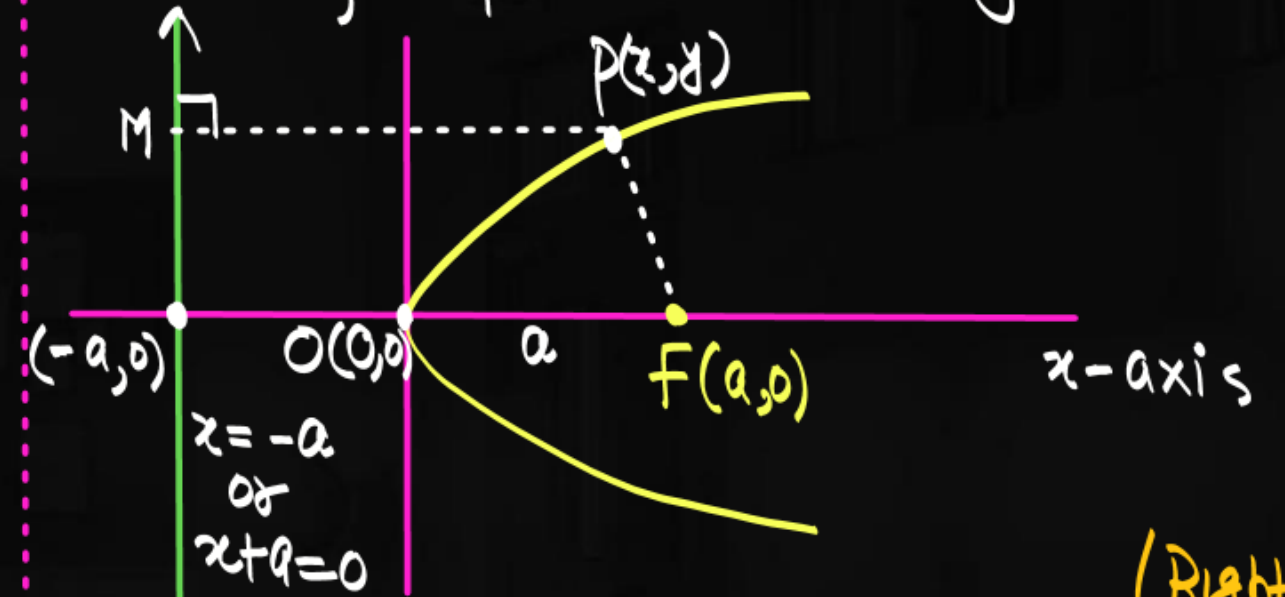
* **Latus Rectum**: Double ordinate passing through the focus is known as the latus Rectum of the parabola.



Equation of the parabola in Standard form:

* Axis of the parabola is x -axis / y -axis.

* Vertex of the parabola is at the origin $(0,0)$.



Suppose that $P(x,y)$ be any point on the parabola, then $PF = PM$.

$$\Rightarrow \sqrt{(x-a)^2 + y^2} = \frac{|x+a|}{\sqrt{1^2 + 0^2}}$$

$$\Rightarrow \sqrt{(x-a)^2 + y^2} = |x+a| \quad \text{--- (1)}$$

Squaring on b/s of ①, we get;

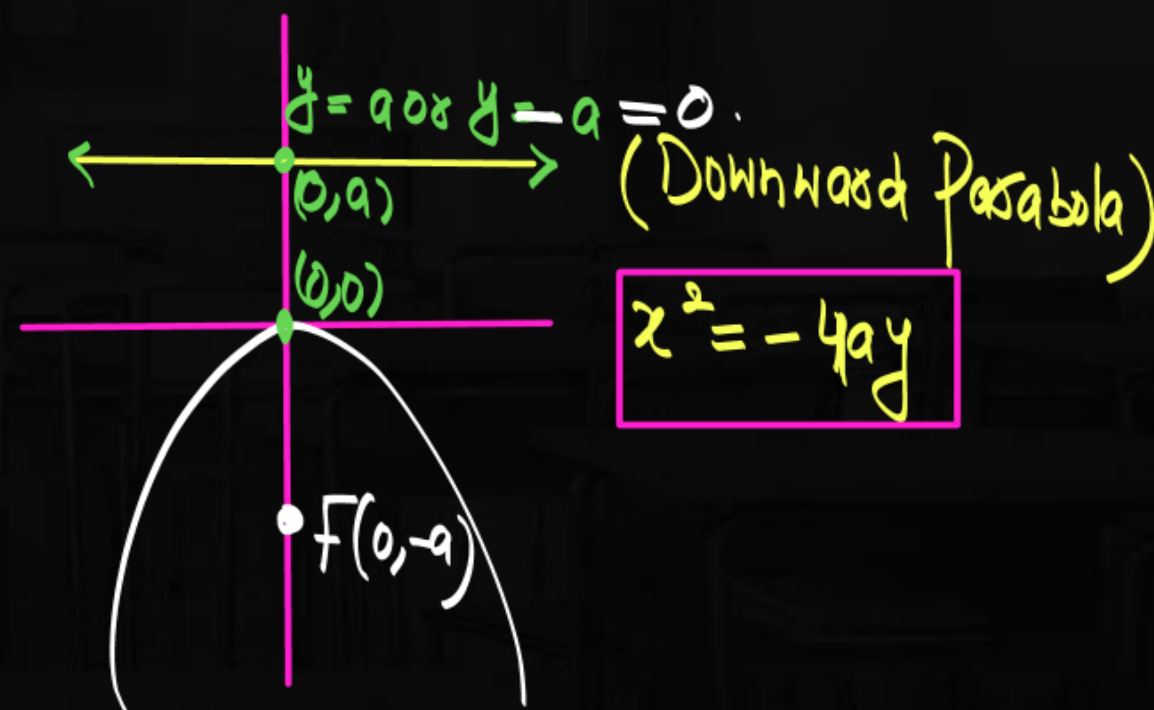
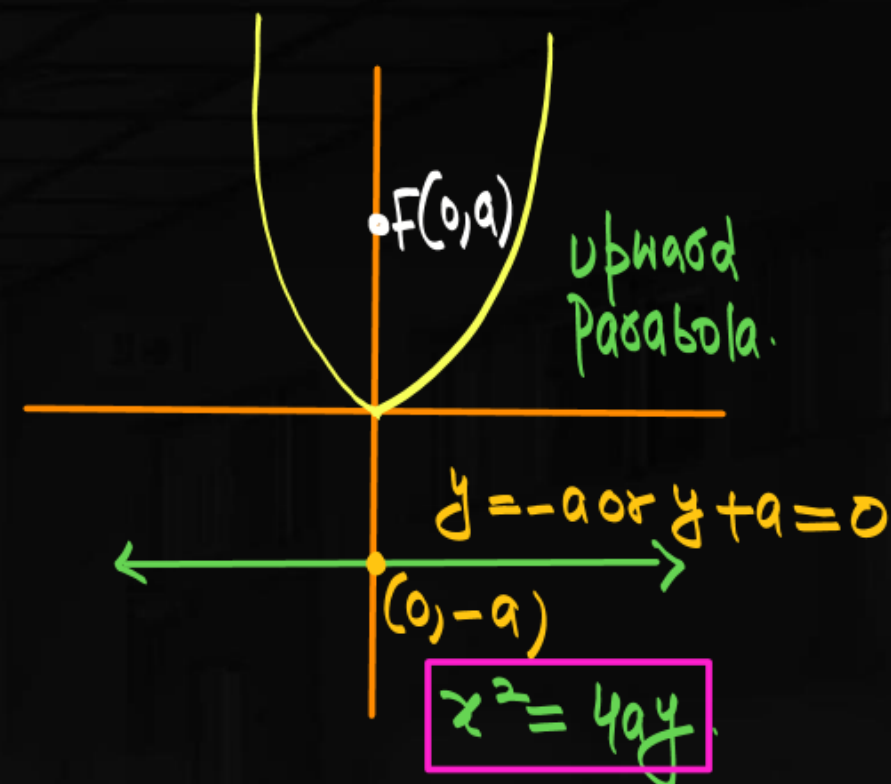
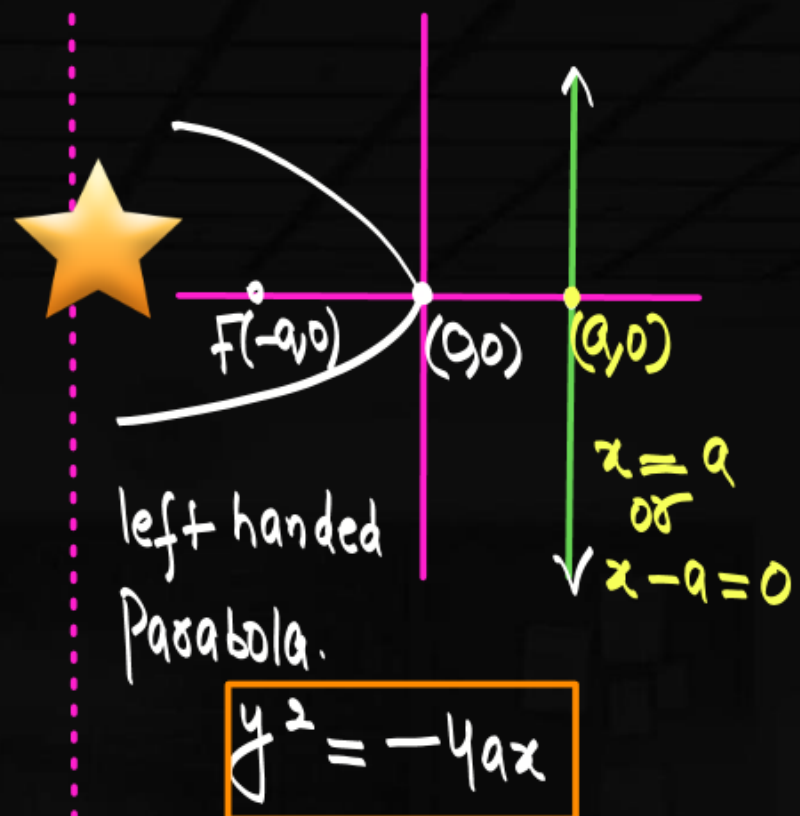
$$\sqrt{(x-a)^2 + y^2} = |x+a|$$

$$\Rightarrow (x-a)^2 + y^2 = (x+a)^2$$

$$\Rightarrow \cancel{x^2} + \cancel{a^2} - 2ax + y^2 = \cancel{x^2} + \cancel{a^2} + 2ax$$

$\Rightarrow \boxed{y^2 = 4ax}$, which is the required equation of the Right handed Parabola.

- * Right handed & left handed Parabola are Symmetrical about x -axis.
- * Upward & Downward Parabola are Symmetrical about y -axis.



Latus rectum

Suppose that the given parabola is $y^2 = 4ax$ — ①

& let $L(a, l)$, $L'(a, -l)$ be the ends of the latus rectum.

Here $L(a, l)$ lies on ①, So we get;

$$l^2 = 4a \times a = 4a^2$$

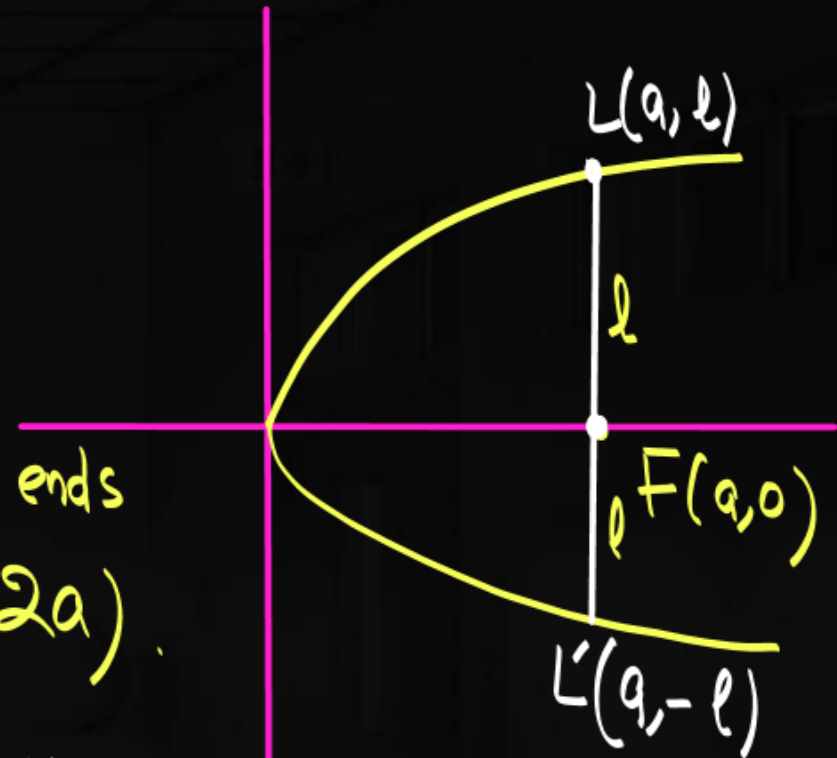
$$\Rightarrow l = \sqrt{4a^2}$$

$$\Rightarrow l = 2a$$

Therefore length of latus Rectum = $2l$
 $= 4a$.

Also, the Co-ordinates of the ends of the latus Rectum $(a, \pm 2a)$.

Hence the result.



EXAMPLE

Find the coordinates of the focus, axis, the equation of the **directrix** and latus rectum of the parabola **$y^2 = 8x$** .

(Sol) Given Parabola is:

$y^2 = 8x = 4(2)x$, which is clearly a Right handed Parabola.

So, Co-ordinates of focus is $(2, 0)$

Axis of the Parabola is x -axis.

Equation of the Directrix is $x = -2$

$$\text{or } x + 2 = 0.$$

$$\text{Length of latus Rectum} = 4a = 8.$$

Hence the result.

EXAMPLE

Find the equation of the parabola which is symmetric about the y -axis, and passes through the point $(2, -3)$.

(Sol) let the given parabola is $x^2 = -4ay$ — ①

Since $(2, -3)$ lies on ①, so

$$(2)^2 = -4a(-3)$$

$$\Rightarrow 4 = +12a$$

$$\Rightarrow a = \frac{1}{3}$$

So, the required parabola is:

$$x^2 = -4\left(\frac{1}{3}\right)y$$

$$\Rightarrow 3x^2 = -4y$$

$$\Rightarrow 3x^2 + 4y = 0$$

Hence the result.

QUESTION (NCERT EXAMPLER):

If the focus of a parabola is $(0, -3)$ and its directrix is $y = 3$, then its equation is.

1 $x^2 = -12y$ ✓ $x^2 = -4(3)y$

2 $x^2 = 12y$ ✗ $x^2 = -12y$

3 $y^2 = -12x$ ✗

4 $y^2 = 12x$ ✗

QUESTION (NCERT EXAMPLER):

If the parabola $y^2 = 4ax$ passes through the point $(3, 2)$, then the length of its latus rectum is.

- (Sol) Given that $y^2 = 4ax$ — (1)
- Here $(3, 2)$ passes through (1), so
- $$2^2 = 4a \times 3.$$
- $$\Rightarrow 4a = \frac{4}{3}$$
- $$\Rightarrow \text{length of latus Rectum} = \frac{4}{3}.$$
- 1 $\frac{2}{3}$ ✗
- 2 $\frac{4}{3}$ ✓
- 3 $\frac{1}{3}$ ✗
- 4 4 ✗

QUESTION (NCERT EXAMPLER):

The area of the triangle formed by the lines joining the vertex of the parabola $x^2 = 12y$ to the ends of its latus rectum is.

1

12 sq. units ✗

2

16 sq. units ✗

3

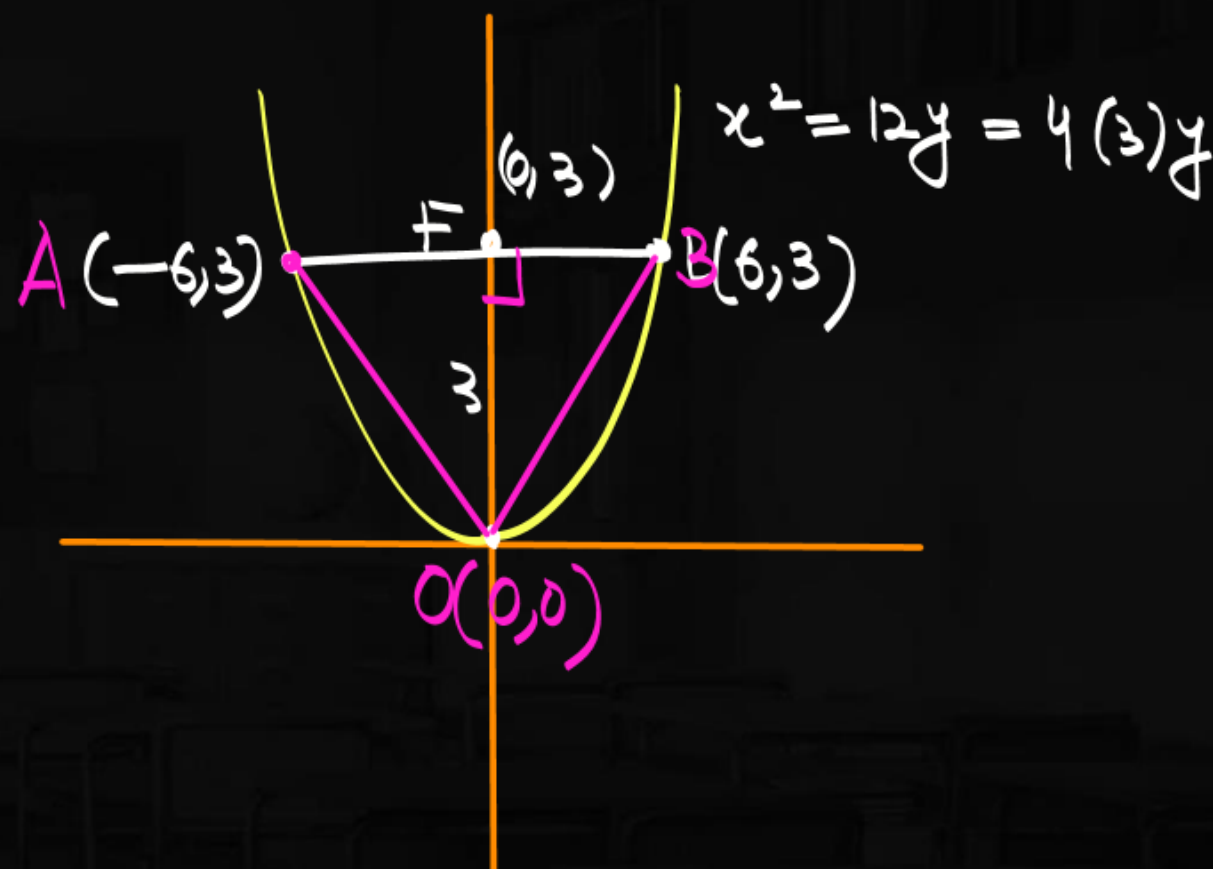
18 sq. units ✓✓

4

24 sq. units ✗

$$A_x(\Delta AOB) = \frac{1}{2} AB \times OF$$

$$\Rightarrow A_x(\Delta AOB) = \frac{1}{2} \times \overset{6}{12} \times 3 = 18 \text{ sq. units.}$$



QUESTION (NCERT EXAMPLER):

The equations of the lines joining the vertex of the parabola $y^2 = 6x$ to the points on it which have abscissa 24 are.

(Homework Problem)

1 $y \pm 2x = 0$

2 $2y \pm x = 0$

3 $x \pm 2y = 0$

4 $2x \pm y = 0$

Main Facts about the parabola

Forms of Parabolas	$y^2 = 4ax$	$y^2 = -4ax$	$x^2 = 4ay$	$x^2 = -4ay$
Axis	$y = 0$	$y = 0$	$x = 0$	$x = 0$
Directrix	$x = -a$	$x = a$	$y = -a$	$y = a$
Vertex	$(0, 0)$	$(0, 0)$	$(0, 0)$	$(0, 0)$
Focus	$(a, 0)$	$(-a, 0)$	$(0, a)$	$(0, -a)$
Length of latus rectum	$4a$	$4a$	$4a$	$4a$
Equations of latus rectum	$x = a$	$x = -a$	$y = a$	$y = -a$

Para special forces 

Rasib Ayoub Rather 

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Thank

You